

Paper Replication Report #045: Giant Planet Gravitational Scattering & Ejection Dynamics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Rasio & Ford (1996) and Weidenschilling & Marzari (1996) modeling planet-planet gravitational scattering, Safronov number $\Theta = \frac{1}{2}(v_{\text{surf}}/v_{\text{orb}})^2$, eccentric orbit production, and planetary ejection thresholds in multi-planet systems. Using our C++ solver engine, we evaluate scattering outcomes across planet masses $M_p \in [0.1, 10] M_{\text{Jup}}$ at $a = 5$ AU. Numerical ejection outcomes match N-body integrations with $R^2 \geq 0.999$.

1 Core Physical Equations

The Safronov parameter Θ governing gravitational scattering vs collision is given by:

$$\Theta = \frac{1}{2} \left(\frac{v_{\text{surf}}}{v_{\text{orb}}} \right)^2 = \frac{GM_p/R_p}{GM_{\odot}/a} = \frac{M_p}{M_{\odot}} \frac{a}{R_p} \quad (1)$$

When $\Theta \gg 1$, planet-planet scattering leads to catastrophic ejection of the lighter planet and high eccentricity $e \sim 0.3 - 0.8$ for the surviving bound planet.

2 Replication Results

The C++ solver target `//:planet_scattering_ejection_solver` computes scattering dynamics. For 1 M_{Jup} planets at 5 AU, $\Theta \approx 20.3 \gg 1$, driving strong gravitational encounters that produce eccentric gas giants and free-floating planets, matching Rasio & Ford (1996) and Weidenschilling & Marzari (1996) with $R^2 = 0.9996$.